

Quiz 18

Multiple-choice questions on thermal expansion

1. When the temperature of a rod of copper is increased, its length:
(a) stays the same (b) increases (c) decreases
2. The amount by which unit length of a material increases when the temperature is raised one degree is called the coefficient of:
(a) cubic expansion (b) superficial expansion (c) linear expansion
3. The symbol used for volumetric expansion is:
(a) γ (b) β (c) L (d) α
4. A material of length L_1 at temperature θ_1 K is subjected to a temperature rise of θ K. The coefficient of linear expansion of the material is α K⁻¹. The material expands by:
(a) $L_2(1 + \alpha\theta)$ (b) $L_1\alpha(\theta - \theta_1)$ (c) $L_1[1 + \alpha(\theta - \theta_1)]$ (d) $L_1\alpha\theta$
5. Some iron has a coefficient of linear expansion of 12×10^{-6} K⁻¹. A 100 mm length of iron piping is heated through 20 K. The pipe extends by:
(a) 0.24 mm (b) 0.024 mm (c) 2.4 mm (d) 0.0024 mm
6. If the coefficient of linear expansion is A , the coefficient of superficial expansion is B and the coefficient of cubic expansion is C , which of the following is false?
(a) $C = 3A$ (b) $A = B/2$ (c) $B = \frac{3}{2}C$ (d) $A = C/3$
7. The length of a 100 mm bar of metal increases by 0.3 mm when subjected to a temperature rise of 100 K. The coefficient of linear expansion of the metal is:
(a) 3×10^{-3} K⁻¹ (b) 3×10^{-4} K⁻¹ (c) 3×10^{-5} K⁻¹ (d) 3×10^{-6} K⁻¹
8. A liquid has a volume V_1 at temperature θ_1 . The temperature is increased to θ_2 . If γ is the coefficient of cubic expansion, the increase in volume is given by:
(a) $V_1\gamma(\theta_2 - \theta_1)$ (b) $V_1\gamma\theta_2$ (c) $V_1 + V_1\gamma\theta_2$ (d) $V_1[1 + \gamma(\theta_2 - \theta_1)]$
9. Which of the following statements is false?
(a) Gaps need to be left in lengths of railway lines to prevent buckling in hot weather
(b) Bimetallic strips are used in thermostats, a thermostat being a temperature- operated switch
(c) As the temperature of water is decreased from 4°C to 0°C contraction occurs
(d) A change of temperature of 15°C is equivalent to a change of temperature of 15 K

10. The volume of a rectangular block of iron at a temperature t_1 is V_1 . The temperature is raised to t_2 and the volume increases to V_2 . If the coefficient of linear expansion of iron is α , then volume V_1 is given by:

(a) $V_2[1 + \alpha(t_2 - t_1)]$ (b) $\frac{V_2}{1 + 3\alpha(t_2 - t_1)}$

(c) $3V_2\alpha(t_2 - t_1)$ (d) $\frac{1 + \alpha(t_2 - t_1)}{V_2}$

Answers

1. (b) 2. (c) 3. (a) 4. (d) 5. (b)
6. (c) 7. (c) 8. (a) 9. (c) 10. (b)